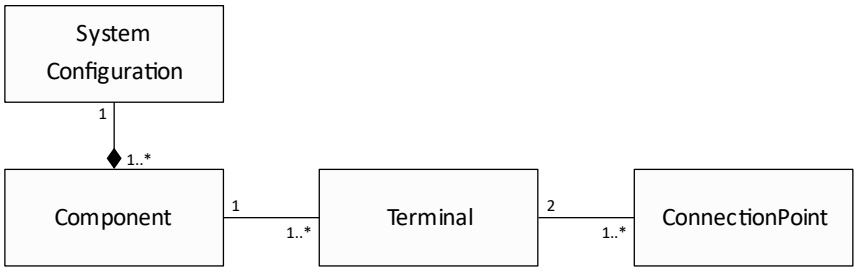
This document constitutes a reference description form to facilitate and set a standard for information exchange between stakeholders on the definition of a system configuration (SC). The SC is the "static" part of a system description. Even though dynamic attributes (e.g., transient parameters) may be included, it does not contain information about the use of the system.

The SC is a detailed technical description of a system and its inherent properties. Formally, an SC comprises a set of components. These components are associated to terminals, which define how they interact with other components. The composition of components to systems is achieved by specifying the connections between these terminals.



NOTE: This document has been carefully structured in a way that allows to extract the contained information in an automated way and to translate it to other machine-readable formats (parsing and conversion by script). Therefore, when editing this document, please keep the following in mind:

- *Only information (text and figures) provided in the cells of the tables defined below will be extracted.*
- *The first cell of each table contains a specific title that identifies its purpose ("Document History", "System Configuration Identification", etc.). Do NOT change these titles, even if you replicate a table (as in the case for the element descriptions).*
- *When extending tables (e.g., add more instances to an element description), insert new row with the same layout. Especially do NOT split or merge cells in a table's row.*
- *You may add additional (sub-)headings to structure the document to accommodate your needs.*
- *Names and identifiers should be machine-readable, i.e., no spaces, no special symbols, only basic ASCII letters and numbers.*

1. Document History

Please keep track of significant changes to this document. Consider using the following versioning scheme to indicate the stage of the SC: 0.x – concept stage, 1.x – test design stage, 2.x – post test / documentation stage, 3.x and higher – revisiting the test.

Document History			
Date	Version	Description	Author

2. General description

2.1 System configuration identification

Please specify a name for this SC together with an ID to identify it.

NOTE: Please use a machine-readable ID.

System Configuration Identification	
System configuration ID	Name

2.2 Short description of context

Please provide a short description of the SC's context in the table below. This will help readers understand what the SC is intended for and for what kind of test cases it relevant. Optionally, an ontology / namespace associated with the component and variable naming may be referenced.

Short Description of Context
Context description
Key figures
Keywords
Ontology / namespace (optional)

3. System Breakdown Diagram (SBD)

Please add in the table cell below the System Breakdown Diagram (SBD), which gives an overview of all types of elements that constitute the SC. The elements are represented as classes. They are organized on different branches of a tree to reflect different domains and they are related to each other on different "levels" (of detail). There exist two types of "vertical" relations between a level (n) class and a level (n+1) class. Their representation in the SBD is compliant with UML notations:

- *Containment relation: a class "has a" sub-class. Containment relations are depicted with a filled diamond (on the side of the container class) and a solid line (example: a rotor is part of a generator).*
- *Inheritance relation: a sub-class "is a" subtype of its parent class. Inheritance relations are depicted with an unfilled triangle (on the side of the parent class) and a solid line. Inheritance relations suppose a transfer of the descriptive parameters from the parent class to the inherited class (example: an underground cable is a type of electric interconnection).*

Every class in the SBD should have an alphanumeric tag, to which it can be referred to in the list of SC elements further down below.

For further details please refer to this document (p. 13ff).

NOTE: Please use only machine-readable names for class names and tags.

System Breakdown Diagram
<add figure here>

4. Graphical representations of SC

Please provide domain-specific graphical representations of the SC in the table cells below. This may include network diagrams or any other technical or non-technical drawings.

NOTE: Please put each drawing in a separate table cell. Each table cell with a drawing should be preceded by a table cell containing the drawing's title.

Graphical Representation of System Configuration
Overview
<add figure here>
Detailed view (optional)
<add additional figures here, replicate this and the preceding cell if required>

5. Terminal descriptions

In the SBD only "vertical" relations between classes (containment or inheritance) are depicted. "Horizontal" connections to further specify interfaces between objects (instances of classes) and relations between them (energy and/or information exchanges) are defined via terminals.

NOTE: Please use only machine-readable names for type IDs.

Terminal Description				
Name	Type ID	Type of exchange	Connected component classes	Comment

6. Component descriptions

Please replicate the table below for each of the elements described in the SBD. This table provides detailed information an element, for instance about its functionality and attributes, its interfaces to other elements, or its instances that are part of the SC.

NOTE: Please use only machine-readable names for IDs, tags, attribute names, etc.

Component description	
Tag in SBD	
Level in SBD	
Class ID	
Parent class	

Contained in		
Description		
Number of components in SC		
Attributes		
Functionality		
Physical characteristics		
Terminals		
Terminal ID	Type ID	Description
Instances		
Instance ID	Instance parameterization	

7. Connection points

Please list the connection points between all the component instances contained in the SC in the table below.

Connection points			
Connection from component	Connection from terminal	Connection to component	Connection to terminal